

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

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1 Introduction

Every year, billions of products enter the European market with sustainability labels, yet consumers have little reliable way of checking whether claims such as “low carbon” or “ethically sourced” are accurate. A study commissioned by the European Commission found that more than half of the green environmental declarations lack verifiable evidence to support, eroding public confidence in corporate environmental commitments [32].

To address this credibility gap, the Ecodesign for Sustainable Products Regulation (ESPR) now mandates Digital Product Passports (DPPs) —standardised lifecycle records—for most EU products [9]. With the implementation of compliance in 2027, reliable DPP solutions have become a market urgent need[9].

Our product EcoPassEU is a blockchain-based DPP platform which allows manufacturers to record product information as permanent, publicly verifiable entries on the Ethereum blockchain. Unlike traditional databases, the immutability of blockchain records means that once the data is released, it cannot be altered retrospectively, eliminating the possibility of post-hoc falsification [7]. Any user can obtain a complete and encrypted product history by entering the product ID or scanning the QR code.

EcoPassEU serves four key stakeholders. It provides SME manufacturers with a cost-effective ESPR compliance channel through a web for batch data entry and automated QR code generation. At the same time, consumers, regulators and retailers can also get open and wallet-free access. Just search for the product ID to immediately obtain the blockchain-verified product passport and supply chain map, enabling the independent verification of environmental claims without prior technical expertise.

The value of EcoPassEU is threefold. Primarily, it provides manufacturers with a practical way to ESPR deadlines, reducing the risk of market exclusion. Meanwhile, the platform directly cracked down on greenwashing, securing product records via cryptographic hashing replaces brand self-reporting with independent, third-party verification. Thirdly, it promotes the development of circular economy by equipping recyclers with the precise material data necessary to recover resources when products are scrapped.

2 Background

2.1 Digital Product Passports and the Regulatory Context

A Digital Product Passport (DPP) is a structured digital record encompassing lifecycle data—such as material composition, provenance, and end-of-life guidance—designed to inform stakeholders and support more sustainable design choices, efficient resource use, and the transition towards circular business models [1,3,6,19].

The regulatory foundation for DPPs comes from the European Green Deal and Circular Economy Action Plan [14,15]. Based on these 2 frameworks, the Ecodesign for Sustainable Products Regulation (ESPR), came into force on 18 July 2024,, mandates DPPs for almost all physical goods sold in the EU, regardless of the seller’s location [9].

2.2 The Problem: Trust in Sustainability Claims

Despite regulatory momentum, the implementations of most DPPs rely on centralised, proprietary databases [30]. This architecture encourages the selective concealment of data, hindering aftermarket participants like independent repairers [10]. Furthermore, managing huge volume of lifecycle data centrally raises significant concerns regarding storage efficiency, data integrity, and vulnerability to breaches [37]. Taken together, downstream actors — including importers, recyclers, and end consumers — lack reliable ways to independently verify product information.

This problem is exacerbated by the prevalence of greenwashing, broadly understood as false, vague, or unsubstantiated environmental claims that make products appear more sustainable than they are [23]. A 2020–2021 EU sweep of 344 online claims found that nearly half were potentially deceptive—often substituting verifiable details with vague terms like “conscious”—and 59% lacked accessible evidence [32]. Another EU website screening found that 42% of sustainability claims were misleading enough to constitute improper commercial behaviour [32].

Blockchain provides a structural solution to this trust problem. Because records on a public blockchain are replicated across a distributed network, neither single party can unilaterally tamper with the records, providing a form of tamper evidence that cannot be replicated by a centralised system [7,27]. Smart contract introduced in the Ethereum protocol allows data access and update rules to be encoded in publicly readable logic, so that there is no need for trusted intermediaries and enabling cross-organisational verification without shared infrastructure [39].

2.3 Related Work and Existing Platforms

Blockchain-based product traceability uses its immutability and decentralised verification to improve supply chain transparency [7]. Blockchain-based product traceability emerged as a research topic around 2015–2016, with early work

concentrated in food and agriculture before spreading to pharmaceuticals, electronics, and textiles [11,34]. More recently, research has shifted from feasibility towards scale and efficiency: for instance, Walmart’s use of IBM Food Trust — built on Hyperledger Fabric — to reduce food-item tracing time from six days to under three seconds [11] .

In the commercial field, Circularise is among the closest comparable platforms, offering DPPs with selective disclosure of sustainability data across supply chains, with applications spanning the automotive, battery, and textile sectors [8]. VeChain similarly deploys blockchain and smart contracts on its VeChainThor platform to achieve supply chain transparency, though it runs on a permissioned network with a restricted set of authority-approved validators [36]. Both platforms have a common limitation: reliance on proprietary infrastructure or permissioned networks introduces vendor dependency and reduces interoperability [4,38].

EcoPassEU takes a different approach. Rather than relying on a proprietary database, it is built on the public Ethereum blockchain and IPFS, with all rules encoded in open smart contracts. Product data is stored on IPFS, whilst only a small set of cryptographic identifiers is written on-chain, keeping costs low. The batch registration function further reduces the cost per product, making the platform viable for SMEs. And as an open-source project, it carries no vendor lock-in risk.

2.4 Business Viability

EcoPassEU’s revenue model rests on two foundations. The first is public funding: the EU’s CIRPASS-2 project (2024–2027) explicitly subsidise DPP deployment across textiles, electronics, and construction, with dedicated support for SMEs adopting “DPP-as-a-Service” models [16]. As an open source platform, EcoPassEU is fully capable of benefitting from such grants.

Another is subscription revenue. ESPR compliance is legally mandatory for any business selling physical goods in the EU, forming a captive market independent of voluntary commitments [9]. The green technology and sustainability market is projected to grow from USD 25.47 billion in 2025 to USD 73.90 billion by 2030 [25]. EcoPassEU charges a graded subscription fee based on passport volume, with optional premium tiers for API integration and compliance reporting. The platform is built on public Ethereum, supporting payment with ETH or other ERC-20 tokens, providing international users a lower-friction settlement option and removing the need for traditional banking intermediaries.

3 Design Documentation

3.1 Design Philosophy and Architectural Choices

EcoPassEU is a DPP designed to deliver ESPR-compliant Digital Product Passports with independently verifiable sustainability data. The design is governed by three design principles, each addressing a specific product requirement.

First, a hybrid storage model is adopted: full product data is pinned to IPFS via Pinata [5], whilst only the content identifier (CID), a `keccak256` integrity hash, the issuer address, and a timestamp are recorded on-chain [39], keeping gas costs low. Second, commercially sensitive data remains off-chain by design; the on-chain hash serves as an integrity anchor, confirming that the off-chain document is unaltered without exposing its contents to the ledger. Third, all read queries use a public RPC endpoint, allowing consumers and regulators to inspect any passport without holding a wallet or incurring gas fees.

3.2 System Architecture

Figure 1 shows the four-layer architecture of the overall system

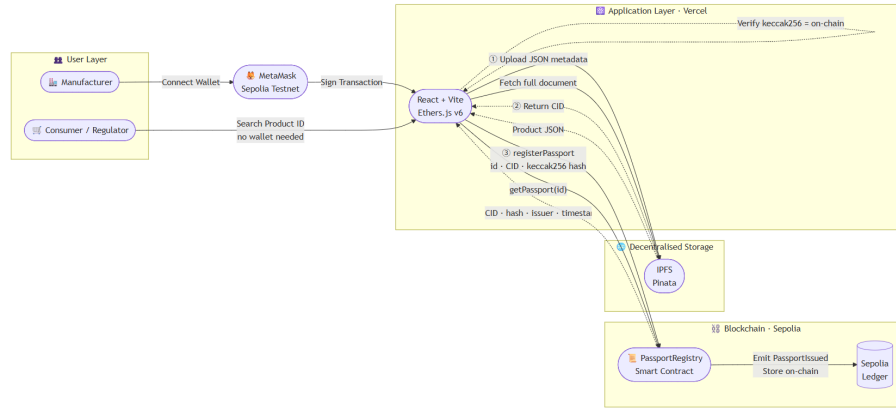


Fig. 1. EcoPassEU system architecture.

At the user layer, two distinct actors participate in the system: manufacturers, who connect through MetaMask to sign and submit transactions, and consumers or regulators, who can query product records without a wallet. Both routes pass through the application layer — a React/Vitefrontend deployed on Vercel, using Ethers.js v6 for contract calls — which is responsible for uploading documents to IPFS, calling the Sepolia test network and local integrity check. The `PassportRegistry` smart contract records the CID, metadata hash, issuer address, and timestamp on the Sepolia ledger, emitting a `PassportIssued` event upon registration. An `exists` flag can prevent duplicate entries. Supply chain provenance is visualised via Leaflet.js with OpenStreetMap.

3.3 Use Cases and Workflow

EcoPassEU serves three actor types with differing technical needs, as illustrated in Figure 2.

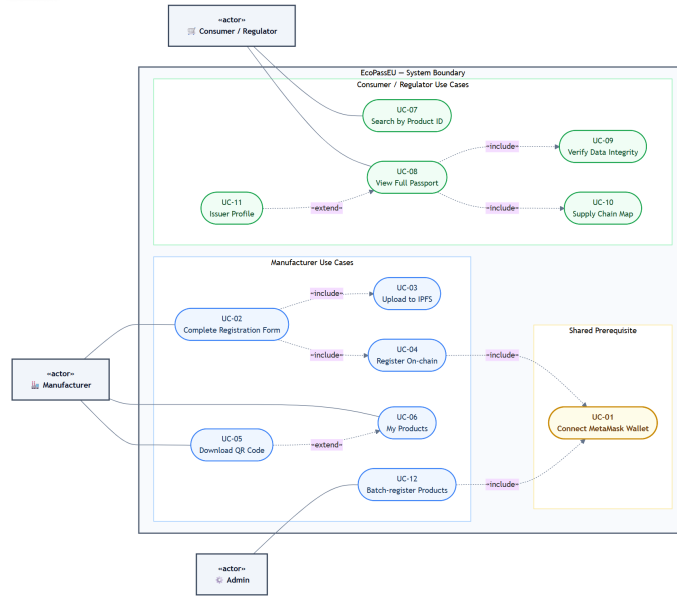


Fig. 2. EcoPassEU use case diagram.

Wallet connection via MetaMask on the Sepolia testnet (UC-01) is a common premise for all manufacturers and administrators, as each on-chain write must be signed by a wallet key pair and cryptographically linked to an accountable issuer [20,21]. To register a product, the manufacturer fills in a structured registration form (UC-02), including product details, material composition, and supply chain sources. Upon submission, the frontend serialises the form data to JSON and uploads it to IPFS via Pinata (UC-03) to receive a CID.

Next, the system calculates the `keccak256` hash of the JSON locally and passes it to the `registerPassport()` function on the smart contract alongside the CID and product ID (UC-04). After the user confirms the transaction in MetaMask, a `PassportIssued` event is emitted, and a QR code with an Ether-scan link is provided for users to download (UC-05), which also accesses the product list view (UC-06). The admin role requires the same wallet connection but adds batch registration capabilities (UC-12) to reduce the unit gas cost for high-volume producers.

On the read side, consumers and regulators require no wallet to verify a product. The user submits a product ID (UC-07), and the frontend queries the contract via a public RPC endpoint to jump to the full passport view (UC-08). If the passport exists, the system obtains the complete JSON data from IPFS and recalculates its hash on the client: a green integrity badge is displayed if it matches, and a warning if it does not match (UC-09). The product passport page will also display an interactive supply chain map (UC-10) and an optional issuer profile (UC-11). This separation between the write-gated and read-open flows

supports the design goal of providing transparent information without setting financial or technical barriers to end users.

3.4 Sequence and State Diagrams

Figure 3 outlines the technical details behind the core application flows, focussing on the strict execution order required for successful registration. Specifically, the frontend must first obtain a valid CID from IPFS before prompting the wallet signature and triggering the smart contract. This ensures that the on-chain pointer never leads to data loss. Meanwhile, the verification flow demonstrates how the system relies entirely on the client logic to compare the acquired IPFS documents against the on-chain hash, ensuring trustless validation.

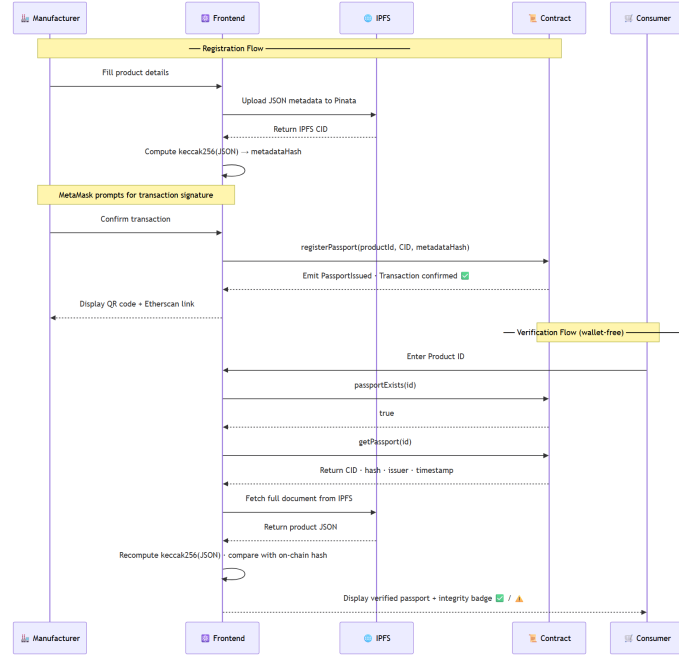


Fig. 3. UML sequence diagram detailing the registration and wallet-free verification flows.

Figure 4 presents an UML state machine diagram to track the digital passport’s lifecycle. Physical products initially entered the system as **Unregistered**. Only after the data is successfully uploaded to Pinata will the status change to **IPFSStored**. Following the user’s signature, the status changes to **TxPending**, and then after mining the block on Sepolia, it reaches the immutable **Registered**. During the final verification stage, the client hash comparison will mark the record as either **Verified** (if the hash matches) or **Compromised** (if the data has been altered).

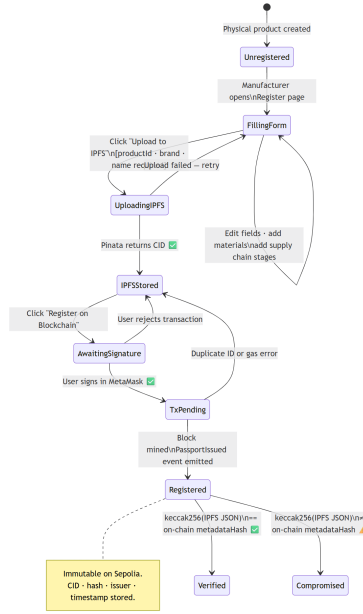


Fig. 4. UML state machine diagram showing the lifecycle of a digital product passport.

3.5 User Interface Design

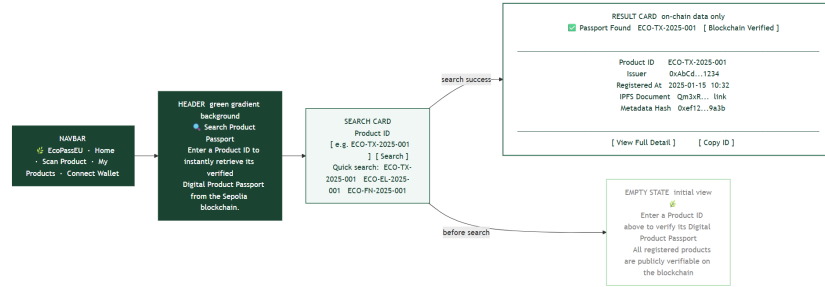


Fig. 5. Wireframe of the search interface, showing the initial state and the on-chain result card.

To reflect the environmental theme, the page adopts a dark green colour scheme (**#2d6a4f**, **#1b4332**). The layout uses standard **rem** spacing for a clean and consistent visual effect. As shown in Figure 5, the search interface is straightforward. Users can query a product ID via the central search bar or use the pre-defined quick search buttons, entirely without connecting a wallet. The interface transitions from an initial empty state to a verified result card upon a successful search. This card immediately displays the core on-chain data—such as the

issuer address, registration timestamp, and metadata hash—providing instant verification before the user proceeds to view the full product passport.

Figure 8 displays the registration layout. A simple tracker (Fill Details → Upload IPFS → Register Chain → Done) shows the user’s registration progress. All form fields—product details, materials, and supply chain—are displayed on a scrollable page instead of hiding sections. If MetaMask is not connected, an amber warning banner will be displayed to stop users from uploading data until they log in.

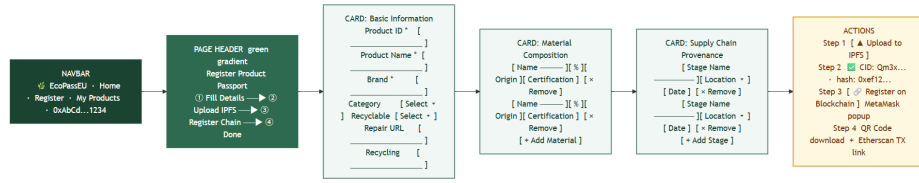


Fig. 6. Wireframe layout of the manufacturer registration form.

As seen in Figure 9, the result page clearly divides the information. The *Blockchain Record* card displays on-chain data (issuer, time, and hash), while other cards display IPFS details. Integrity badge—a green check for a match, or an amber triangle for an error—makes it easy to judge whether the data is valid. If a user does not install MetaMask, the page will automatically use public RPC to load information.



Fig. 7. Wireframe detailing the structure of the product verification page.

Our site is built with CSS Grid and Flexbox to run perfectly on any device. At 600px, forms will switch to a single-column layout to adapt to the mobile device screen. A 768px breakpoint adjusts the menu and interactive map sizes for tablet users.

4 Project Execution

4.1 Core Code Analysis and Data Structures

EcoPassEU uses a hybrid architecture, separating on-chain and off-chain data to balance transparency with efficiency. The core data structure is divided across these two environments.

Off-chain, the complete lifecycle data—including category, brand, material composition, and supply chain provenance—is structured as JSON objects and

pinned to IPFS via Pinata. On-chain, the `PassportRegistry.sol` smart contract uses a strictly typed `Passport` struct to store only the essential, immutable references:

Listing 1.1. Core Data Structure in `PassportRegistry.sol`

```
struct Passport {
    string ipfsCID;           // off-chain
                             document pointer
    bytes32 metadataHash;    // keccak256
                             of JSON for tamper-evidence
    address issuer;
    uint256 timestamp;
    bool exists;
}

// productId => Passport
mapping(string => Passport) private passports
;
```

This private mapping directly associates a unique product ID with its corresponding passport data for $O(1)$ retrieval. Initially, we considered using ERC-1155 tokens to represent these passports; however, we opted for this lightweight registry approach. Storing only the IPFS CID, a cryptographic hash, the issuer address, and a block timestamp on-chain significantly reduces deployment complexity and gas costs per registration, whilst keeping the full product lifecycle data off-chain on IPFS.

Listing 1.2 outlines the core registration functions and reflects a number of optimisations for Solidity:

Listing 1.2. `registerPassport` function

```
function registerPassport(
    string calldata productId,
    string calldata ipfsCID,
    bytes32 metadataHash
) external {
    require(!passports[productId].exists,
        "Passport already exists");
    passports[productId] = Passport({
        ipfsCID: ipfsCID,
        metadataHash: metadataHash,
        issuer: msg.sender,
        timestamp: block.timestamp
    },
        exists: true
    );
    emit PassportIssued(productId,
        ipfsCID, metadataHash,
        msg.sender, block.timestamp);
}
```

Passing string parameters as `calldata` instead of `memory` avoids redundant data copying and significantly reduces the cost of gas. Meanwhile, the `external` visibility modifier allows direct argument access from the call data, providing greater gas efficiency compared with a `public` declaration. Whats more, binding the `issuer` directly to `msg.sender`—a secure EVM primitive—guarantees unforgeable cryptographic attribution for every passport, bypassing the need for supplementary access controls.

As a supplement to the registration logic, the `PassportIssued` event will be issued after each successful execution:

Listing 1.3. PassportIssued Event Definition

```
event PassportIssued(
    string indexed productId,
    string indexed ipfsCID,
    bytes32 metadataHash,
    address indexed issuer,
    uint256 timestamp
);
```

By designating `productId` and `issuer` as `indexed`, EVM stores these parameters in the Bloom filter of the transaction receipt. This mechanism allows off-chain clients to efficiently filter logs by specific product or manufacturer without scanning the computing overhead of each block, which represents best-practice model of establishing queryable audit tracking in Ethereum DApps [39].

4.2 Testing Strategy

To validate the reliability of `PassportRegistry`, a unit-testing suite was implemented using the Hardhat environment alongside the Chai assertion library [?], following established practices for smart contract validation [?]. Listing 1.4 shows the shared test fixture; a fresh contract instance is deployed via `beforeEach`, ensuring full isolation between cases.

Listing 1.4. Hardhat test fixture – shared setup

```
beforeEach(async function () {
    [owner, addr1] = await hre.ethers.
        getSigners();
    const Factory =
        await hre.ethers.getContractFactory("
            PassportRegistry");
    contract = await Factory.deploy();
});
```

The most critical tests concern `registerPassport`, as shown in Listing 1.5. The first case confirms that `msg.sender` is persisted as the issuer, which is the foundation of the platform’s unforgeable attribution model. The second verifies that attempting to re-register an existing product ID triggers the `require` guard and reverts with the expected reason string, guaranteeing passport immutability.

Listing 1.5. Unit tests – registerPassport: issuer storage and duplicate guard

```

it("should store the correct issuer address",
  async function () {
    await contract.registerPassport(
      PRODUCT_ID, IPFS_CID, META_HASH);
    const [, , issuer] = await contract.
      getPassport(PRODUCT_ID);
    expect(issuer).to.equal(owner.address
    );
  });

it("should revert on duplicate product ID",
  async function () {
    await contract.registerPassport(
      PRODUCT_ID, IPFS_CID, META_HASH);
    await expect(
      contract.registerPassport(PRODUCT_ID,
        IPFS_CID, META_HASH)
    ).to.be.revertedWith("Passport
      already exists");
  });

```

4.3 Final UI Showcase

Figures 8 and 9 show the deployed EcoPassEU interface. The registration portal guides manufacturers through a structured form with a step indicator, whilst the public verification page displays the passport details, integrity status, and IPFS document link without requiring a wallet.

Fig. 8. Deployed EcoPassEU interface: manufacturer registration portal.

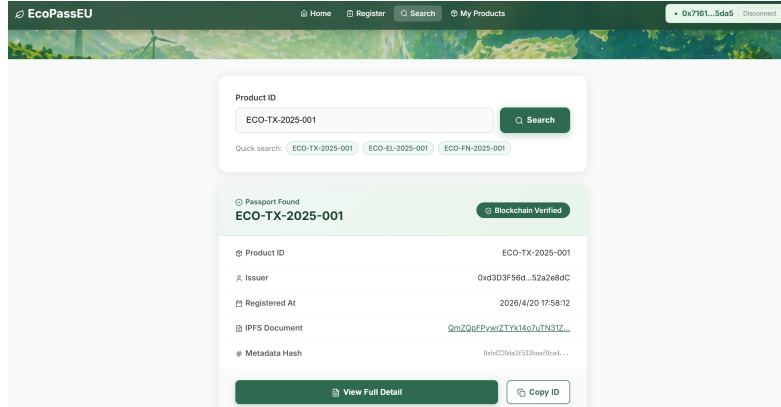


Fig. 9. Deployed EcoPassEU interface: public product verification page.

4.4 User Manual

The application is designed to be intuitive for two distinct user groups.

For Manufacturers (Registration):

1. Connect your MetaMask wallet to the Sepolia testnet using the top navigation bar.
2. Navigate to the 'Register' page and complete the product detail forms, including material and supply chain data.
3. Click 'Upload to IPFS'. The system will generate a unique CID.
4. Click 'Register on Blockchain' and approve the transaction in the MetaMask pop-up.
5. Once confirmed, download the generated QR code to attach to your physical products.

For Consumers and Regulators (Verification):

1. No wallet connection is required. Open the EcoPassEU homepage.
2. Enter the Product ID into the central search bar, or scan the product's QR code.
3. Review the product's lifecycle data. A green 'Blockchain Verified' badge indicates that the IPFS data matches the immutable blockchain record.

5 Critical Evaluation

A critical assessment of the DApp's execution costs demonstrates highly optimised gas efficiency. The gas cost can be rigorously decomposed using standard EVM opcode pricing [39]. Each registration incurs a base transaction cost of 21,000 gas, plus approximately 20,000 gas per `SSTORE` opcode for each new storage slot written. By keeping heavy string data (e.g., repair guides) off-chain on

IPFS, the on-chain state changes are kept extremely lightweight. Crucially, the `Passport` struct writes five fields, but using `bytes32` for `metadataHash` rather than a hex string halves the storage cost for that specific field, as it fits neatly into a single 32-byte EVM storage word. According to our testnet deployment, executing a single registration transaction consumes approximately 125,430 gas. However, the `batchRegisterPassports` function shows significant operational scalability. By traversing an array of products in one transaction, the contract amortises the fixed 21,000-gas base cost across multiple registrations. Test data confirms that registering 10 products in bulk reduces the average per-product cost to roughly 92,100 gas—explaining the 26% per-unit reduction and rendering the platform economically viable for SMEs.

In terms of security, the implementation has successfully reduced the risk of tampering and unauthorised modification, which is a central concern in Ethereum smart contract security [24,35]. In smart contracts, the explicit `require` check provides a strict tamperability detection guarantee by making the registered passport immutable. In addition, the stored metadata hash establishes a cryptographic commitment between the on-chain registry and the off-chain IPFS load. If the malicious actor tries to intercept and tamper with the IPFS document, the cryptographic hash of the tampered document will no longer match the immutable record saved on the chain, thus immediately invalidating the passport. Unlike existing solutions that rely on permissioned nodes or proprietary databases, EcoPassEU achieves enterprise-level privacy protection on a fully open network through the IPFS hash pairing mechanism.

The contract structurally avoids several common loopholes in Ethereum. The re-entry attack [24] is not applicable because the contract neither accepts nor transfers Ether, eliminating the external call path used by reentrancy attack. Integer overflow and overflow has been mitigated by Solidity 0.8.x compiler’s built-in checked arithmetic, which will revert when overflow occurs without the Safe-Math library [13]. The `private` visibility of `passports` mapping prevents direct external enumeration; all reading are controlled by the typed `getPassport()` and `passportExists()` view functions, which are declared as `view`, therefore cost zero gas when called off-chain.

Although the decentralised architecture provides strong security, it also brings some runtime restrictions. The DApp performs quickly in on-chain verification and can usually be read in the state of completion in milliseconds. However, the access to the complete product history depends on public IPFS gateways, which sometimes have higher latency than centralised cloud servers. In addition, although the current platform runs efficiently on the Sepolia testnet, if it is migrated to the Ethereum mainnet, users will face volatile gas costs. Future versions of EcoPassEU should be considered for deployment on Layer-2 networks such as Polygon. This can not only maintain EVM compatibility, but also achieve near-instant final confirmation; at the same time, if it is necessary to support secondary market transfer in the future, it can be further reconstructed into a plan based on ERC-1155 tokens [31].

6 Conclusion

This project successfully built EcoPassEU, a decentralised application (DApp) that provides Digital Product Passports (DPPs) to meet the EU’s Ecodesign for Sustainable Products Regulation (ESPR) [17]. By using Ethereum smart contracts [2], the system registers and verifies products automatically without relying on central authorities [33]. To prevent false environmental claims, a `keccak256` hash securely links the immutable on-chain record to the full document on IPFS [28]. Running on Ethereum’s Proof-of-Stake network, which reduced energy consumption by approximately 99.95% compared to its predecessor Proof-of-Work consensus [12], directly reinforces EcoPassEU’s sustainability mandate: a DPP platform that itself carries a high carbon footprint would undermine the environmental credentials it is designed to certify. This hybrid storage model also greatly reduces gas fees by keeping heavy data off the blockchain [22].

Technically, the platform is live on the Sepolia testnet. We used Hardhat to test the smart contracts thoroughly, ensuring core functions and edge cases perform exactly as expected [29]. To help small and medium enterprises (SMEs), we added a batch registration feature that significantly cuts transaction costs. The front-end allows consumers to view product histories without a crypto wallet, fetching data smoothly via public nodes [26].

However, the system has current limitations. As a testnet deployment, real-world gas costs and network limits remain unproven. Using a single IPFS provider (Pinata) also introduces a central point of failure. Furthermore, the platform lacks identity verification (KYC), meaning any user can register a product ID. Finally, the strict immutability of the smart contract means honest mistakes cannot be fixed, and products cannot be officially recalled or revoked.

Future work will focus on mainnet deployment and gas profiling. We plan to add decentralised identifiers (DIDs) or verifiable credentials to confirm manufacturer identities, alongside a system to update or revoke passports safely. Moving to a multi-node storage solution like Filecoin will remove the reliance on one IPFS gateway. Finally, zero-knowledge proofs could be explored to hide sensitive supply chain details while still proving compliance to the public. Overall, EcoPassEU proves that blockchain offers a practical, transparent way to track sustainable goods, providing a strong base for future real-world use.

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A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Deployment

- **Smart Contract Address(es):** [Insert Sepolia testnet addresses here]
- **Etherscan Links:** [Insert links to verified contracts on Etherscan]

A.2 Project Links

- **Deployed Website:** [Insert URL]
- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11

A.3 Transaction Hashes

- **Deployment TX:** [Insert Hash] - Explanation.

A.4 Equity Shares & Contributions

- Luxiao Cao: [Weight]
- Akshansh Rajora: [Weight]
- Fuyu Cao: [Weight]
- Tianwei Yu: [Weight]
- Detai Dong: [Weight]

[Insert Screenshot of GitHub - Insights - Contributor here]

A.5 JIRA Project Board

[Insert screenshots of 'Summary', 'timeline', and 'backlog' pages here]

A.6 Individual Reflection Essays

Luxiao Cao [Insert 500-word reflection here]

Akshansh Rajora [Insert 500-word reflection here]

Fuyu Cao [Insert 500-word reflection here]

Tianwei Yu [Insert 500-word reflection here]

Detai Dong [Insert 500-word reflection here]

A.7 Team Sprint Reports

[Insert chronological sprint reports here]

A.8 Complete Source Code

Smart Contracts

```
// Insert complete Solidity code here
```

Frontend Core Files

```
// Insert modified/relevant frontend files (e.g., page.js) here
```